On Geometric Bipartite Graphs with Asymptotically Smallest Zarankiewicz Numbers

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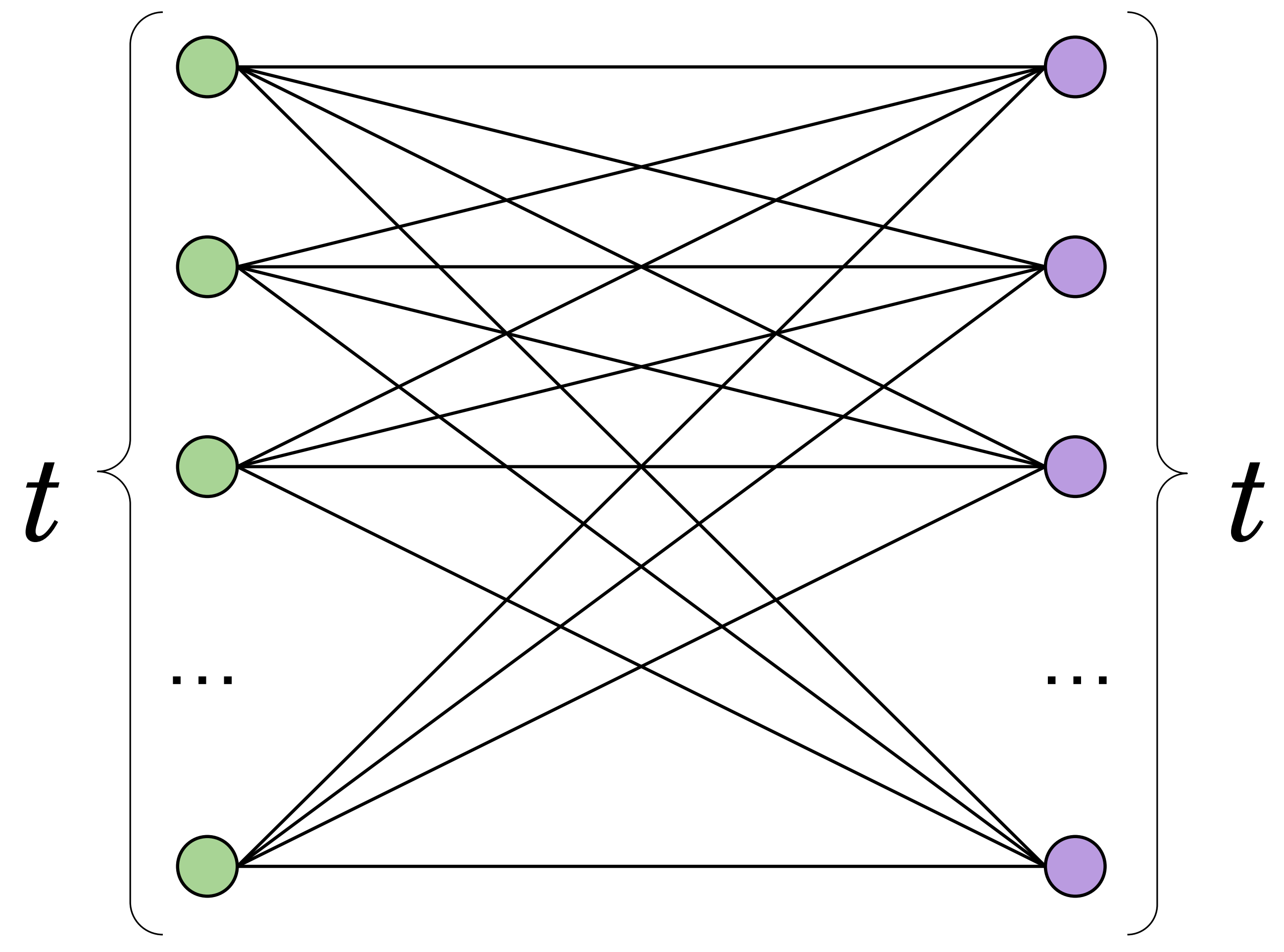
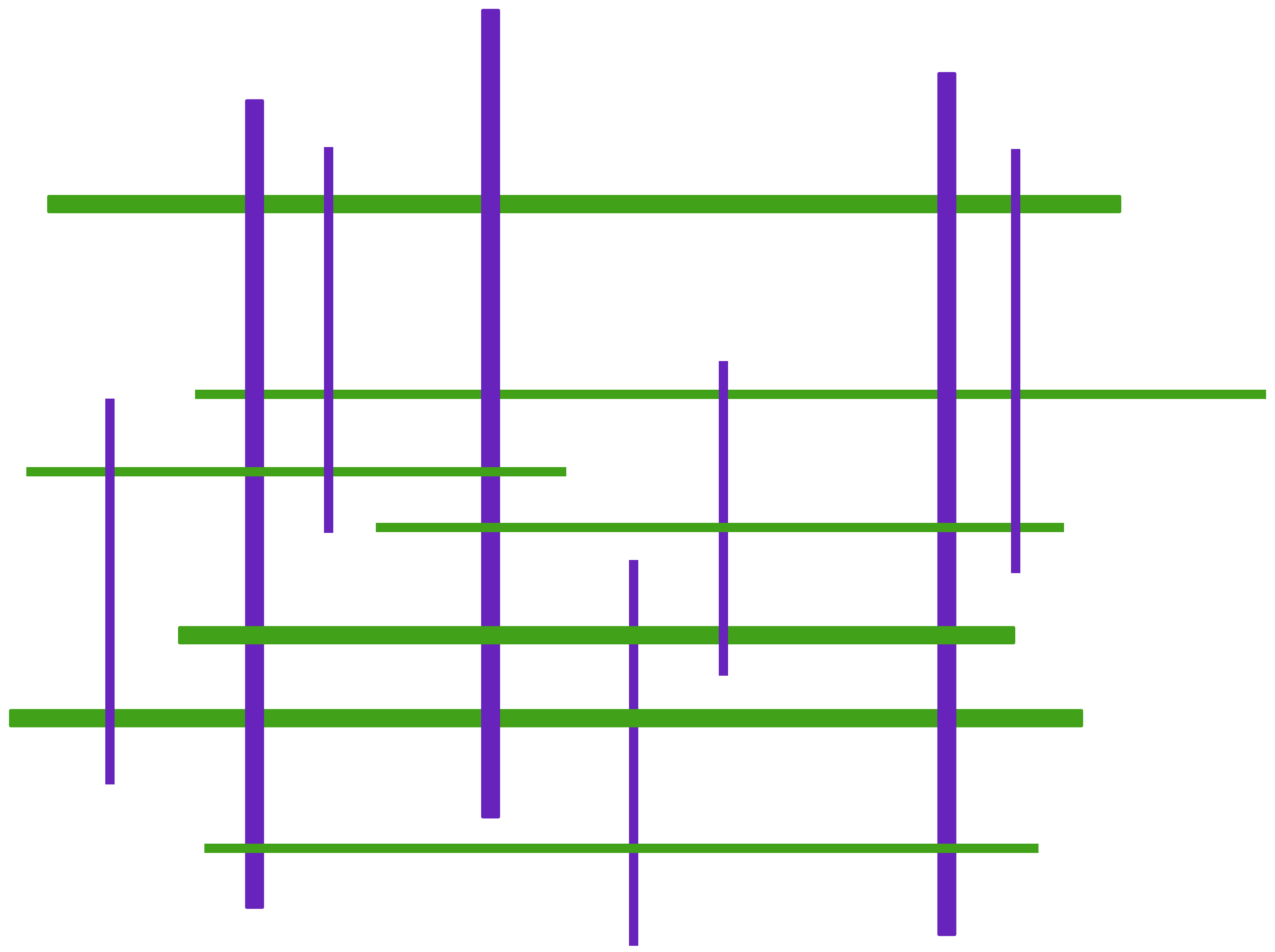
Problem Statement

What is the maximum number of edges in a bipartite graph G in class $\mathscr{C}$, if G does not have $K_{t+1,t+1}$ as a subgraph?



$K_{t,t}$

<u>When $\mathscr{C}$ is grid intersection graphs (GIG):</u> What is the maximum number of intersections between horizontal and vertical segments, without forming a $(t+1)$-by-$(t+1)$ grid?

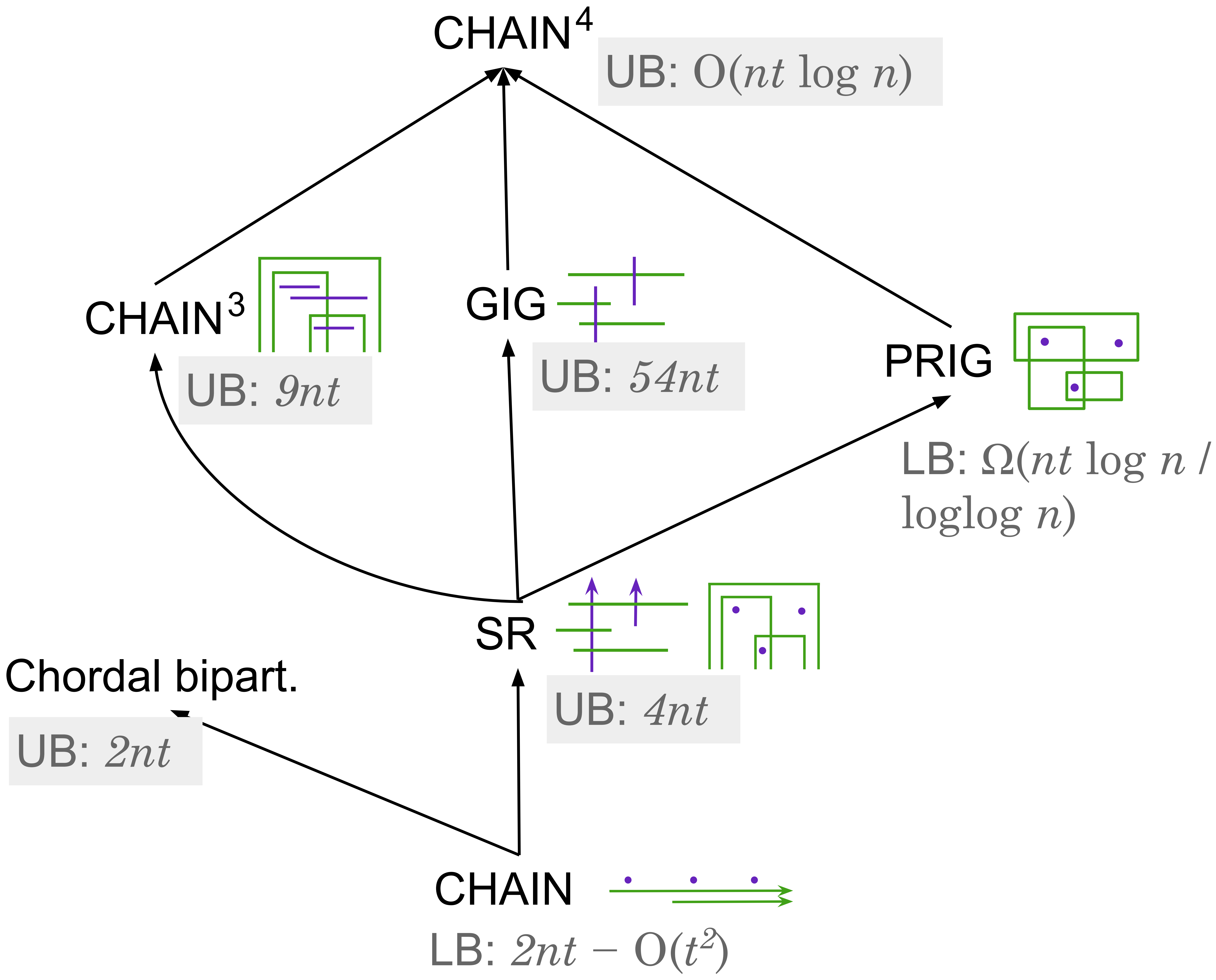


We bound this as a function of t and the number of nodes n in each partition.

Results

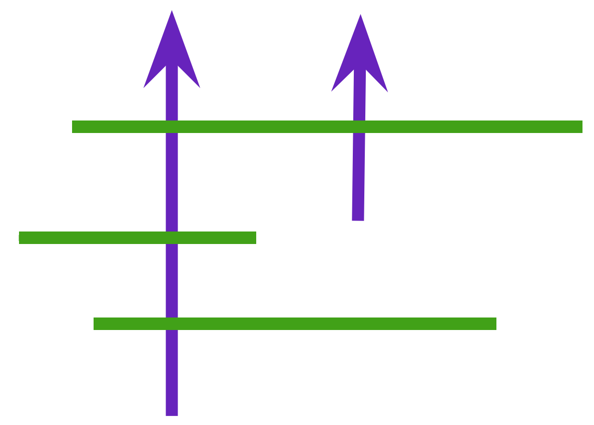
For two graphs, their intersection is defined as the intersection of their node and edge sets.

We use CHAIN^d to denote the class of graphs that are the intersection of d CHAIN graphs.



Upper Bound in SR Graphs

A SR graph G can be drawn as an intersection graph of horizontal segments and upwards rays.



Lemma: If $G \in$ SR does not include $K_{t+1,t+1}$ as a subgraph, then there exists a node of degree at most $2t$.

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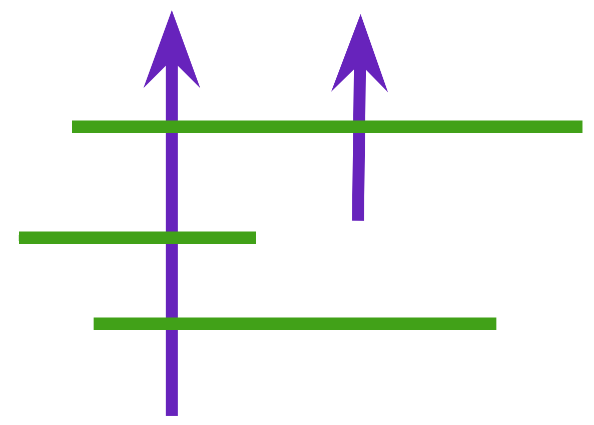

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Proof: Suppose by contradiction that every node in G has a degree at least $2t+1$.

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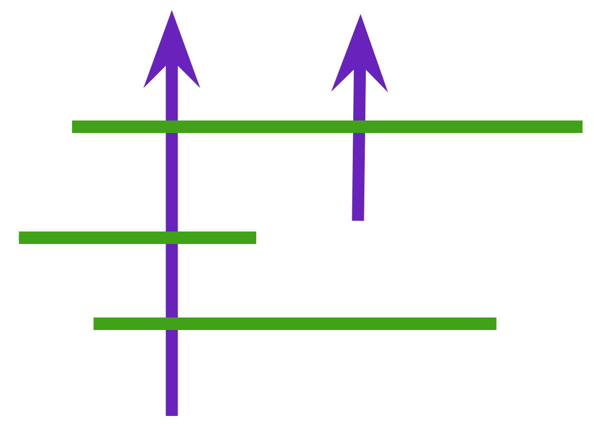

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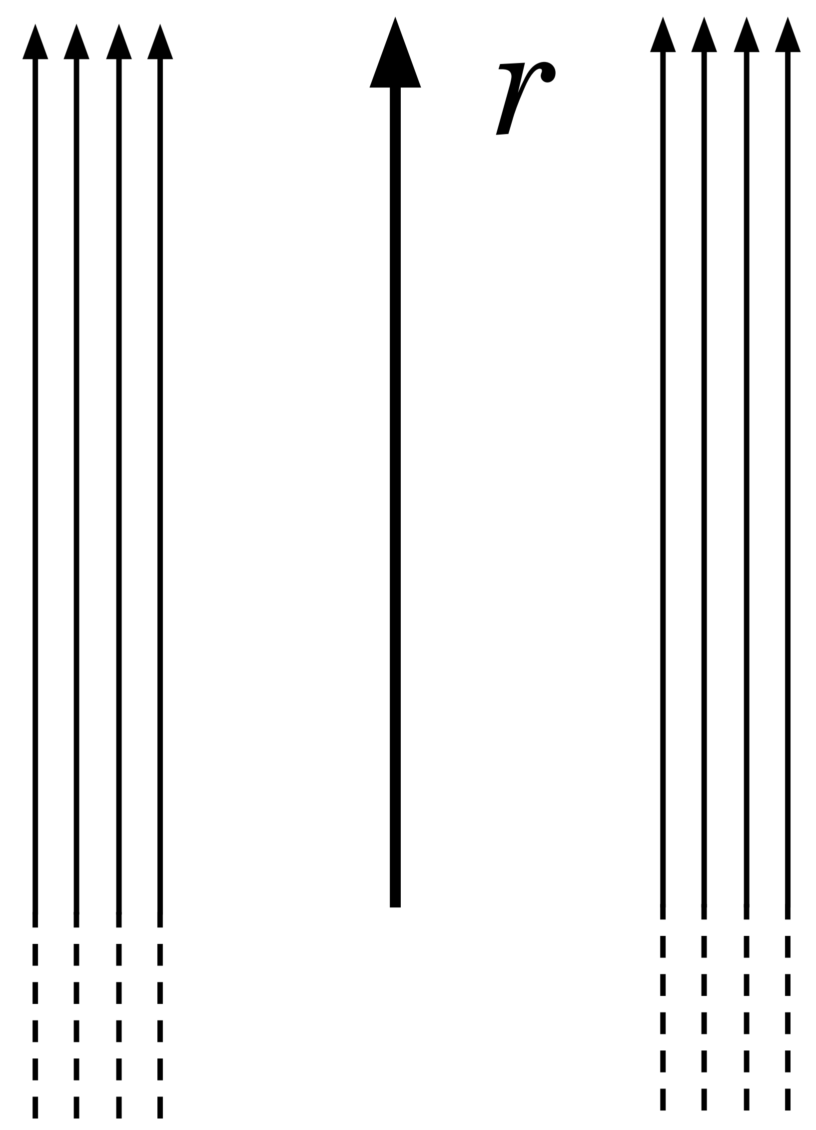

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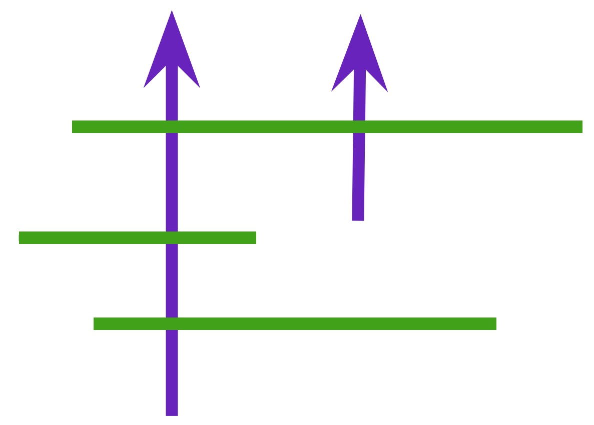

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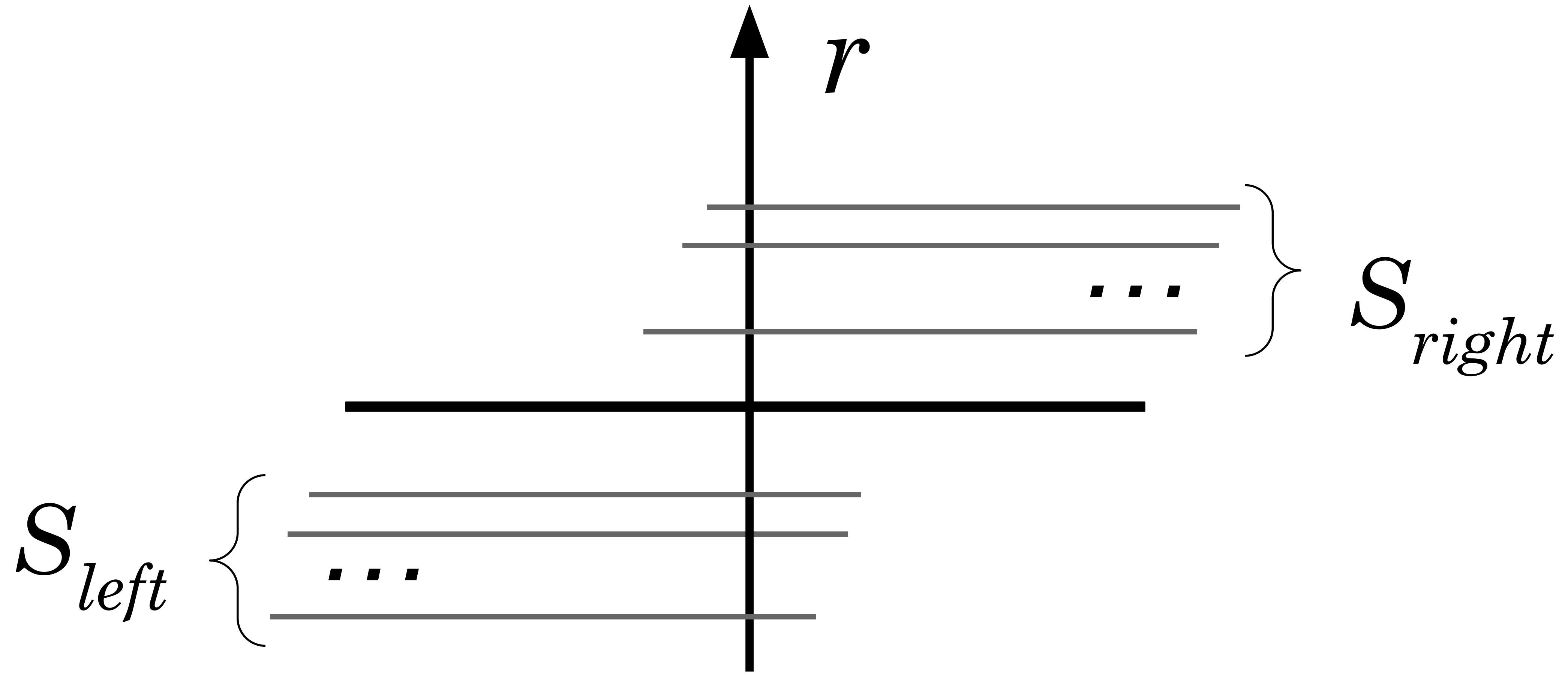

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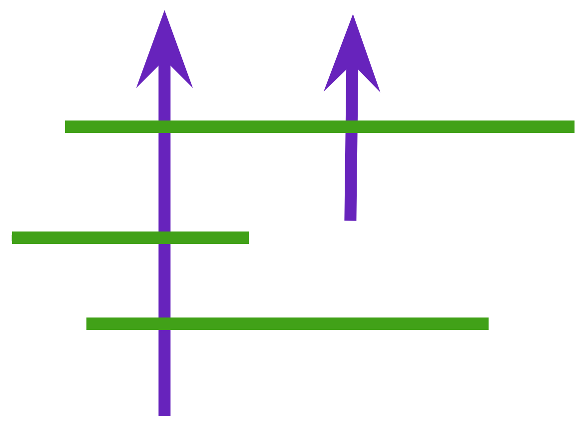

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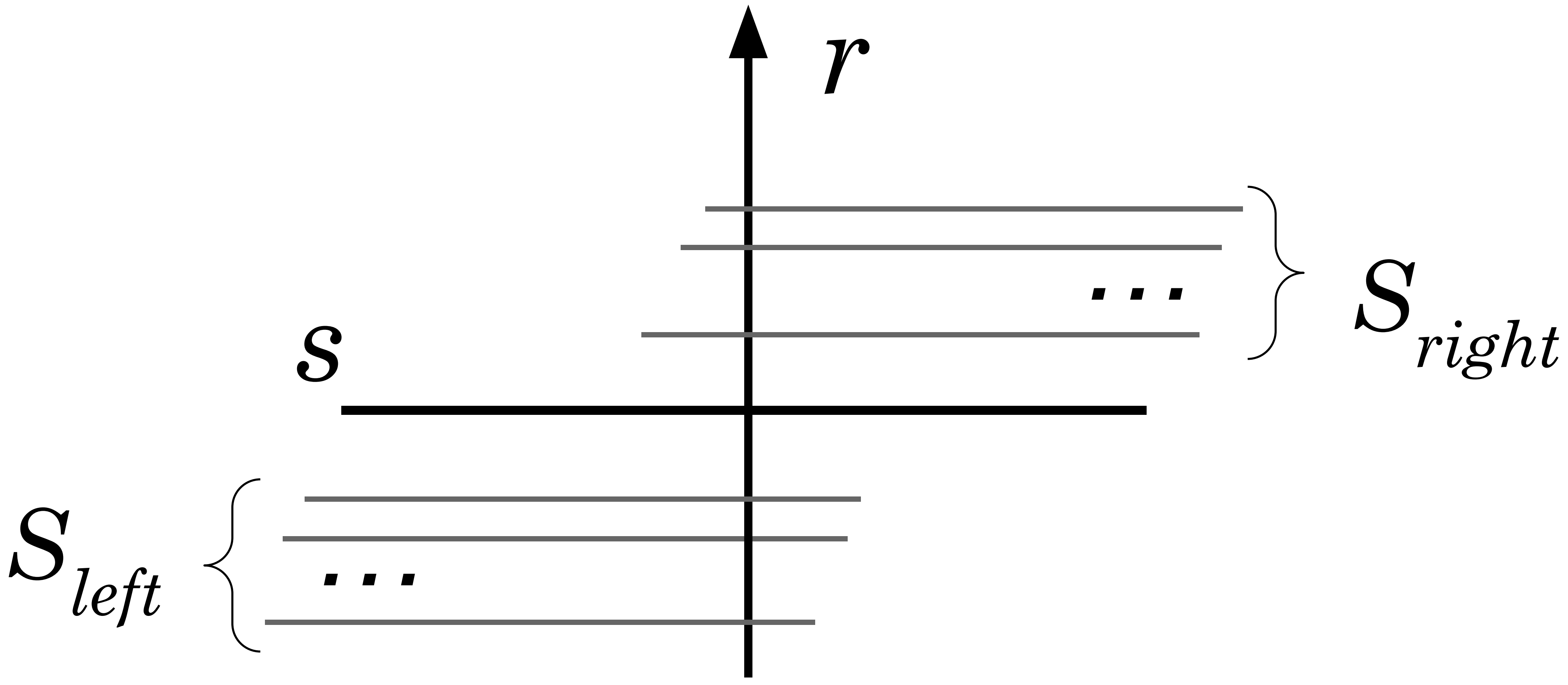

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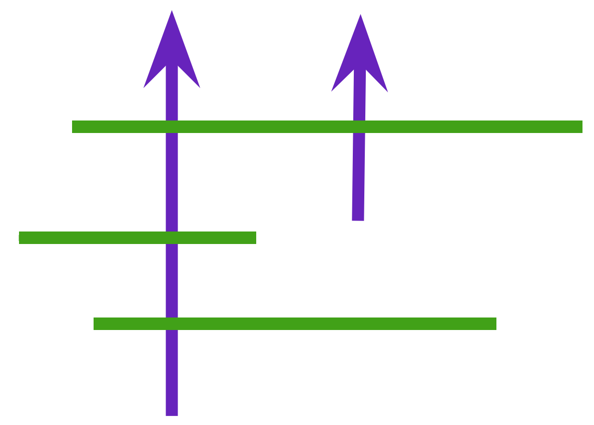

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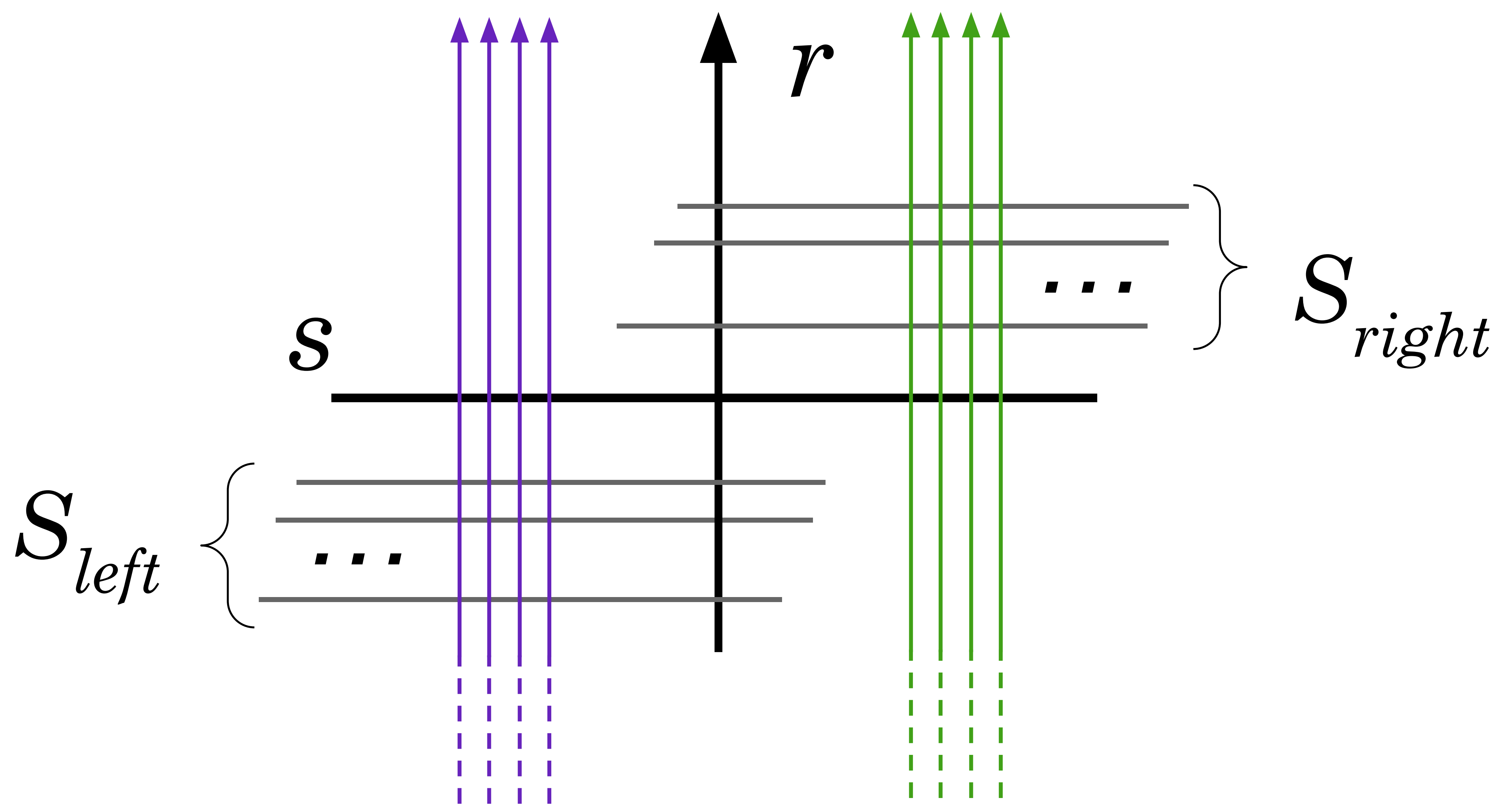

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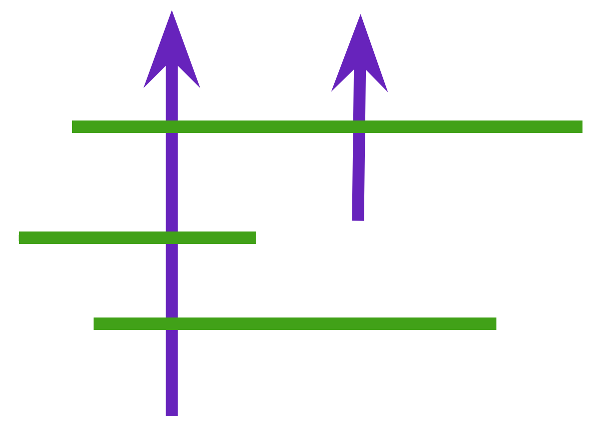
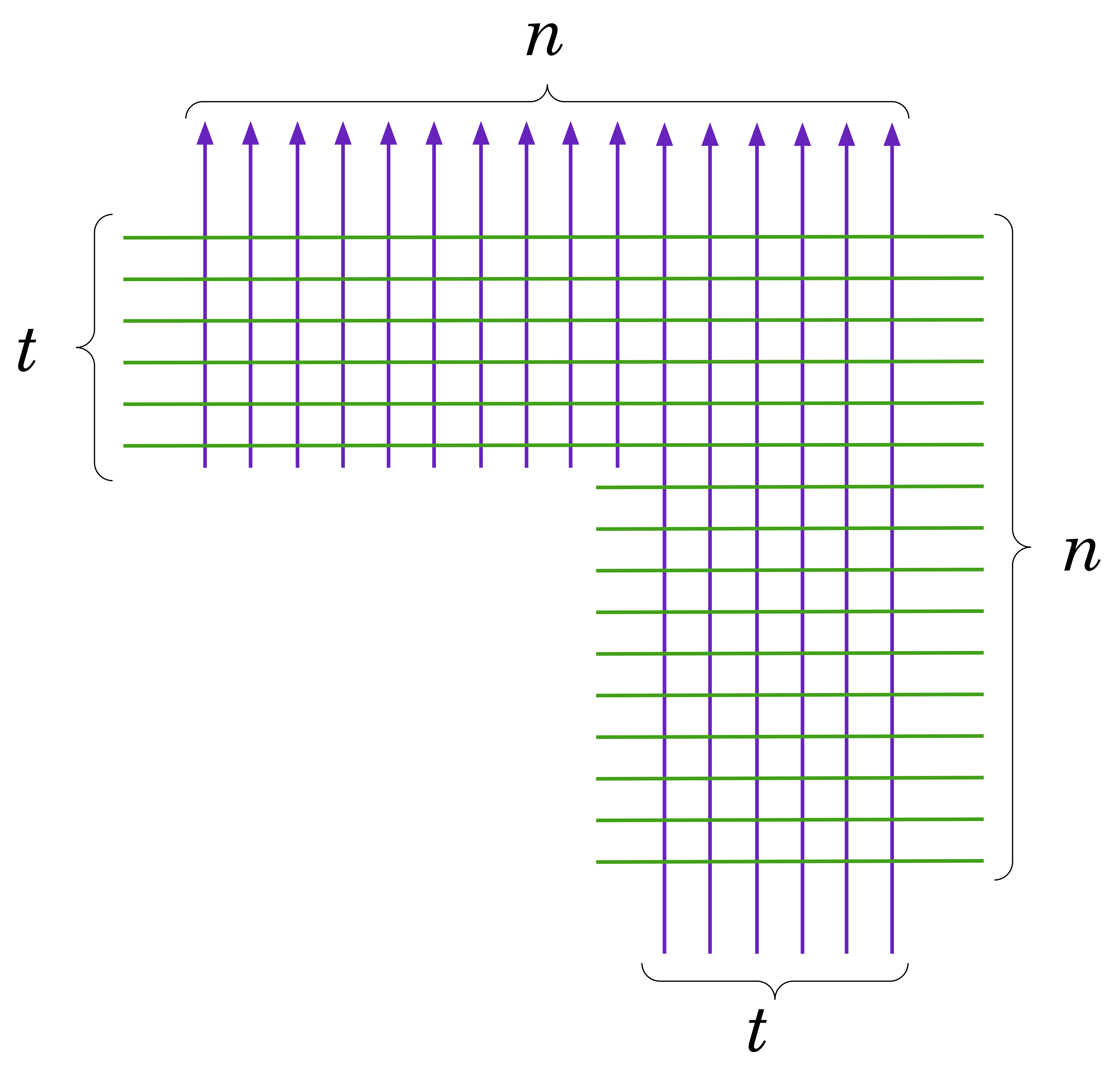


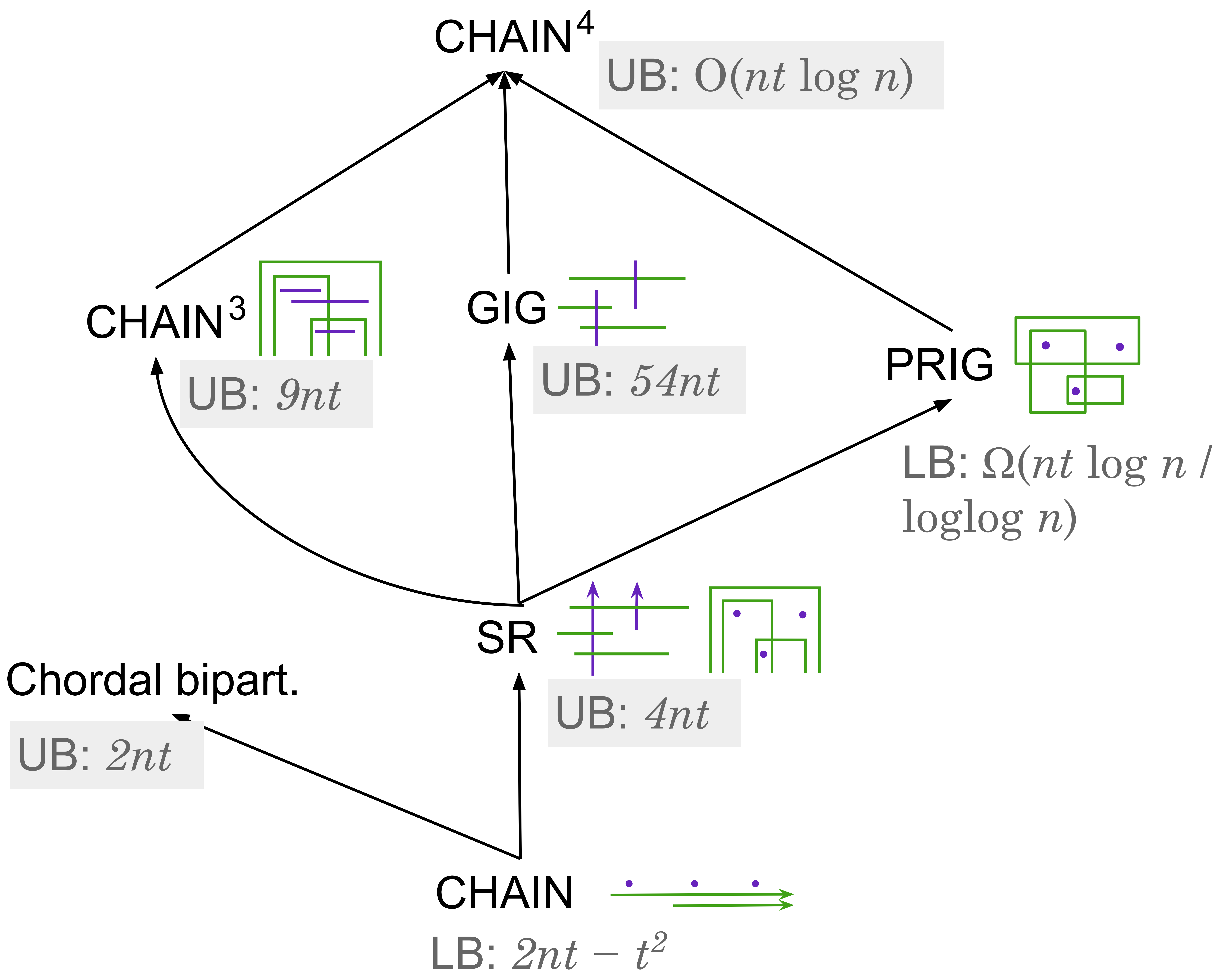
Lemma: If $G \in$ SR does not include $K_{t+1,t+1}$ as a subgraph, then there exists a node of degree at most $2t$.

If G that has n nodes in each partition, has at most $4nt$ edges.

The Lower Bound as a SR graph

A graph $G \in$ SR that does not include $K_{t+1,t+1}$ as a subgraph and has n nodes in each partition can have at least $2nt - t^2$ edges.




CHAIN^4
UB: $O(nt \log n)$
CHAIN^3
UB: $9nt$
GIG
UB: $54nt$
PRIG
LB: $\Omega(nt \log n /$
$\log\log n)$
SR
UB: $4nt$
Chordal bipart.
UB: $2nt$
CHAIN
LB: $2nt - t^2$

Thank You!

Questions?